

Spectre Figure 4.2 / 5.1 Straight-Path Rule Data

Positioned entry-to-exit records for the nine supertiles

Scope. This packet is the extracted *straight-line baseline*: paths crossing each Figure 5.1 supertile between opposing super-edges without internal bends. Bent paths are intentionally excluded and may be added as a later rule-data layer.

Positioned-state convention. A state $X_{n,m}$ enters supertile X on directional super-edge n through its m -th ordinary edge segment, counted 1-based along the oriented super-edge. Across a glued super-edge of length $|E|$, positions match by $m' = |E| - m + 1$.

Figure 4.2 signed-edge dictionary

Each row is the cyclic signed-edge word for one parent hexagon in the extracted local order.

tile	cyclic signed-edge word
Γ	$\alpha^- \alpha^+ \gamma^- \delta^- \beta^+ \beta^-$
Δ	$\gamma^+ \beta^+ \varepsilon^- \alpha^+ \gamma^- \zeta^-$
Θ	$\gamma^+ \beta^+ \theta^+ \beta^+ \eta \beta^-$
Λ	$\gamma^+ \beta^+ \varepsilon^- \alpha^+ \theta^- \beta^-$
Ξ	$\alpha^- \varepsilon^+ \theta^+ \beta^+ \eta \beta^-$
Π	$\alpha^- \varepsilon^+ \varepsilon^- \alpha^+ \theta^- \beta^-$
Σ	$\zeta^+ \beta^+ \varepsilon^- \alpha^+ \gamma^- \delta^+$
Φ	$\gamma^+ \beta^+ \varepsilon^- \varepsilon^+ \eta \beta^-$
Ψ	$\alpha^- \varepsilon^+ \varepsilon^- \varepsilon^+ \eta \beta^-$

Figure 5.1 straight-path inventory

tile	N	tile	N	tile	N
Γ	8	Δ	10	Θ	6
Λ	8	Ξ	8	Π	10
Σ	8	Φ	8	Ψ	10

Total: 76 positioned straight-path records.

Source excerpts. The build reads only the bundled Figure 4.2 tile-edge macro excerpt and Figure 5.1 supertile excerpt, not the full paper.

Supertile Γ

8 records

state	entry super	entry	word	exit super	exit
$\Gamma_{0,1}$	α^-	$1 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Theta_1$	δ^-	$5 : \theta^+$
$\Gamma_{0,3}$	α^-	$3 : \beta^-$	$\Gamma_1 \Delta_0$	δ^-	$3 : \alpha^+$
$\Gamma_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	γ^-	$3 : \varepsilon^-$
$\Gamma_{2,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Gamma_5 \Phi_1$	α^+	$3 : \varepsilon^-$
$\Gamma_{3,3}$	δ^-	$3 : \alpha^+$	$\Delta_3 \Gamma_4$	α^-	$3 : \beta^-$
$\Gamma_{3,5}$	δ^-	$5 : \theta^+$	$\Theta_4 \Sigma_5 \Phi_0$	α^-	$1 : \varepsilon^+$
$\Gamma_{4,3}$	γ^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Gamma_{5,3}$	α^+	$3 : \varepsilon^-$	$\Phi_4 \Gamma_2 \Pi_0$	β^+	$2 : \alpha^+$

Supertile Δ

10 records

state	entry super	entry	word	exit super	exit
$\Delta_{0,3}$	γ^+	$3 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Phi_1$	α^+	$3 : \varepsilon^-$
$\Delta_{0,5}$	γ^+	$5 : \beta^-$	$\Gamma_1 \Delta_0 \Xi_0$	α^+	$1 : \beta^+$
$\Delta_{1,1}$	ζ^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Delta_{1,3}$	ζ^-	$3 : \theta^+$	$\Xi_4 \Delta_5 \Phi_0$	ε^-	$1 : \varepsilon^+$
$\Delta_{2,3}$	γ^-	$3 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Delta_{3,1}$	α^+	$1 : \beta^+$	$\Xi_3 \Delta_3 \Gamma_4$	γ^+	$5 : \beta^-$
$\Delta_{3,3}$	α^+	$3 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Phi_0$	γ^+	$3 : \varepsilon^+$
$\Delta_{4,1}$	ε^-	$1 : \varepsilon^+$	$\Phi_3 \Delta_2 \Xi_1$	ζ^-	$3 : \theta^+$
$\Delta_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	ζ^-	$1 : \alpha^-$
$\Delta_{5,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Sigma_3 \Delta_1$	γ^-	$3 : \varepsilon^-$

Supertile Θ

6 records

state	entry super	entry	word	exit super	exit
$\Theta_{0,5}$	γ^+	$5 : \beta^-$	$\Gamma_1 \Delta_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Theta_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	θ^+	$5 : \varepsilon^-$
$\Theta_{2,4}$	η	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Theta_{3,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Delta_3 \Gamma_4$	γ^+	$5 : \beta^-$
$\Theta_{4,5}$	θ^+	$5 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Theta_{5,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Sigma_3 \Delta_1$	η	$4 : \varepsilon^-$

Supertile Λ

8 records

state	entry super	entry	word	exit super	exit
$\Lambda_{0,3}$	γ^+	$3 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Phi_1$	α^+	$3 : \varepsilon^-$
$\Lambda_{0,5}$	γ^+	$5 : \beta^-$	$\Gamma_1 \Delta_0 \Xi_0$	α^+	$1 : \beta^+$
$\Lambda_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Lambda_{2,4}$	θ^-	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Lambda_{3,1}$	α^+	$1 : \beta^+$	$\Xi_3 \Delta_3 \Gamma_4$	γ^+	$5 : \beta^-$
$\Lambda_{3,3}$	α^+	$3 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Phi_0$	γ^+	$3 : \varepsilon^+$
$\Lambda_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Lambda_{5,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Sigma_3 \Delta_1$	θ^-	$4 : \varepsilon^-$

Supertile Ξ

8 records

state	entry super	entry	word	exit super	exit
$\Xi_{0,3}$	α^-	$3 : \beta^-$	$\Gamma_1 \Delta_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Xi_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	θ^+	$5 : \varepsilon^-$
$\Xi_{2,2}$	η	$2 : \varepsilon^+$	$\Psi_3 \Gamma_5 \Phi_1$	ε^+	$4 : \varepsilon^-$
$\Xi_{2,4}$	η	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Psi_0$	ε^+	$2 : \varepsilon^+$
$\Xi_{3,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Delta_3 \Gamma_4$	α^-	$3 : \beta^-$
$\Xi_{4,5}$	θ^+	$5 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Xi_{5,2}$	ε^+	$2 : \varepsilon^+$	$\Psi_3 \Sigma_3 \Delta_1$	η	$4 : \varepsilon^-$
$\Xi_{5,4}$	ε^+	$4 : \varepsilon^-$	$\Phi_4 \Gamma_2 \Psi_0$	η	$2 : \varepsilon^+$

Supertile Π

10 records

state	entry super	entry	word	exit super	exit
$\Pi_{0,1}$	α^-	$1 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Phi_1$	α^+	$3 : \varepsilon^-$
$\Pi_{0,3}$	α^-	$3 : \beta^-$	$\Gamma_1 \Delta_0 \Xi_0$	α^+	$1 : \beta^+$
$\Pi_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Pi_{2,2}$	θ^-	$2 : \varepsilon^+$	$\Psi_3 \Gamma_5 \Phi_1$	ε^+	$4 : \varepsilon^-$
$\Pi_{2,4}$	θ^-	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Psi_0$	ε^+	$2 : \varepsilon^+$
$\Pi_{3,1}$	α^+	$1 : \beta^+$	$\Xi_3 \Delta_3 \Gamma_4$	α^-	$3 : \beta^-$
$\Pi_{3,3}$	α^+	$3 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Phi_0$	α^-	$1 : \varepsilon^+$
$\Pi_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Pi_{5,2}$	ε^+	$2 : \varepsilon^+$	$\Psi_3 \Sigma_3 \Delta_1$	θ^-	$4 : \varepsilon^-$
$\Pi_{5,4}$	ε^+	$4 : \varepsilon^-$	$\Phi_4 \Gamma_2 \Psi_0$	θ^-	$2 : \varepsilon^+$

Supertile Σ

8 records

state	entry super	entry	word	exit super	exit
$\Sigma_{0,3}$	ζ^+	$3 : \alpha^+$	$\Lambda_3 \Sigma_2 \Phi_1$	α^+	$3 : \varepsilon^-$
$\Sigma_{1,3}$	δ^+	$3 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Sigma_{1,5}$	δ^+	$5 : \theta^+$	$\Xi_4 \Delta_5 \Phi_0$	ε^-	$1 : \varepsilon^+$
$\Sigma_{2,3}$	γ^-	$3 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Sigma_{3,3}$	α^+	$3 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Lambda_0$	ζ^+	$3 : \alpha^+$
$\Sigma_{4,1}$	ε^-	$1 : \varepsilon^+$	$\Phi_3 \Delta_2 \Xi_1$	δ^+	$5 : \theta^+$
$\Sigma_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	δ^+	$3 : \alpha^-$
$\Sigma_{5,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Sigma_3 \Delta_1$	γ^-	$3 : \varepsilon^-$

Supertile Φ

8 records

state	entry super	entry	word	exit super	exit
$\Phi_{0,3}$	γ^+	$3 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Phi_1$	ε^+	$4 : \varepsilon^-$
$\Phi_{0,5}$	γ^+	$5 : \beta^-$	$\Gamma_1 \Delta_0 \Psi_0$	ε^+	$2 : \varepsilon^+$
$\Phi_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Phi_{2,4}$	η	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Pi_0$	β^+	$2 : \alpha^+$
$\Phi_{3,2}$	ε^+	$2 : \varepsilon^+$	$\Psi_3 \Delta_3 \Gamma_4$	γ^+	$5 : \beta^-$
$\Phi_{3,4}$	ε^+	$4 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Phi_0$	γ^+	$3 : \varepsilon^+$
$\Phi_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Phi_{5,2}$	β^+	$2 : \alpha^+$	$\Pi_3 \Sigma_3 \Delta_1$	η	$4 : \varepsilon^-$

Supertile Ψ

10 records

state	entry super	entry	word	exit super	exit
$\Psi_{0,1}$	α^-	$1 : \varepsilon^+$	$\Phi_3 \Sigma_2 \Phi_1$	ε^+	$4 : \varepsilon^-$
$\Psi_{0,3}$	α^-	$3 : \beta^-$	$\Gamma_1 \Delta_0 \Psi_0$	ε^+	$2 : \varepsilon^+$
$\Psi_{1,1}$	β^-	$1 : \alpha^-$	$\Gamma_0 \Sigma_1$	ε^-	$3 : \varepsilon^-$
$\Psi_{2,2}$	η	$2 : \varepsilon^+$	$\Psi_3 \Gamma_5 \Phi_1$	ε^+	$4 : \varepsilon^-$
$\Psi_{2,4}$	η	$4 : \varepsilon^-$	$\Delta_4 \Sigma_0 \Psi_0$	ε^+	$2 : \varepsilon^+$
$\Psi_{3,2}$	ε^+	$2 : \varepsilon^+$	$\Psi_3 \Delta_3 \Gamma_4$	α^-	$3 : \beta^-$
$\Psi_{3,4}$	ε^+	$4 : \varepsilon^-$	$\Phi_4 \Sigma_5 \Phi_0$	α^-	$1 : \varepsilon^+$
$\Psi_{4,3}$	ε^-	$3 : \varepsilon^-$	$\Sigma_4 \Gamma_3$	β^-	$1 : \alpha^-$
$\Psi_{5,2}$	ε^+	$2 : \varepsilon^+$	$\Psi_3 \Sigma_3 \Delta_1$	η	$4 : \varepsilon^-$
$\Psi_{5,4}$	ε^+	$4 : \varepsilon^-$	$\Phi_4 \Gamma_2 \Psi_0$	η	$2 : \varepsilon^+$